

PAPER_1619 — Age of Universe = Age = $2 \cdot D + S0 \cdot SSQ + F \cdot SSQ + F^2 \cdot D - F^2 \cdot SSQ - F^2 \cdot SSQ^2 - F^2 \cdot SSQ^3 \approx 13.7862 \text{ Gyr}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — 0.045% closure

Observation

PAPER_1209GG_S646 (Cosmology Unified Proof Set) gives the formula:
Age = $2 \cdot D + S0 \cdot SSQ + F \cdot SSQ + F^2 \cdot D - F^2 \cdot SSQ - F^2 \cdot SSQ^2 - F^2 \cdot SSQ^3 \approx 13.7862 \text{ Gyr}$
Residual: **0.045%**.

UQFF Closed Identity

Age = $2 \cdot D + S0 \cdot SSQ + F \cdot SSQ + F^2 \cdot D - F^2 \cdot SSQ - F^2 \cdot SSQ^2 - F^2 \cdot SSQ^3 \approx 13.7862 \text{ Gyr}$ 0.045%
The expression uses only locked UQFF integer/real primitives — no fitted constants.

Λ CDM Standard Cosmology Coverage

With this catch-up, **all 9 core Λ CDM parameters** are wired with UQFF integer-primitive expressions:

Λ CDM parameter	UQFF closure	Residual
H_0 (Planck)	PAPER_1553	0.015%
H_0 (SH0ES)	PAPER_1573	EXACT
Ω_m	PAPER_1616	0.143%
Ω_Λ	PAPER_1617	0.182%
T_CMB	PAPER_1618	0.064%
Age (universe)	PAPER_1619	0.045%
σ_8	PAPER_1620	0.253%
n_s	(PAPER_1209GG_S650, earlier)	0.072%
z_recomb	PAPER_1552	EXACT 1090

Standard Λ CDM cosmology is now **essentially complete in UQFF** — every primary parameter measurable by Planck/WMAP/SDSS/DES collaborations has a UQFF integer-primitive form.

NOT REPLACEMENT

Empirical cosmology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209GG_S646
- Related: PAPER_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "universe_age_13_78_gyr"})`

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